

Donald Moratz

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Research Interests

My research is in international security, with a focus on authoritarian and developing states: how they signal where they stand toward allies and patrons, how foreign powers project influence within them, and how their governments open or close the space for civil society. Significant competition in the international sphere unfolds below the threshold of open conflict — in managed media, diplomatic pressure, and the contestation of civic space — and I study how these information environments shape political behavior and how governments and foreign powers work to control them. I build and validate text-as-data measures of these behaviors over large multilingual news corpora, using transformer models and LLMs, and pair them with quasi-experimental designs and sensitivity analysis to make the strongest causal case the data allow.

Education

- Ph.D. **University of Pennsylvania**, *Political Science*, International Relations, Methods, and Comparative Politics, 2022 – Expected 2027
- M.A. **University of Pennsylvania**, *Statistics*, Concurrent with Ph.D., Expected 2027
- M.A. **Duke University**, *Economics and Political Science*, May 2019
- M.A. **University of St Andrews**, *Economics and International Relations*, June 2013
(Undergrad)

Working Papers

- “The Signaling Gap: State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine” with Sarah Gerstein. 2025. **Job Market Paper.**
LLM-based sentiment analysis of state-owned media in Russia’s near abroad shows that the 2022 invasion shifted coverage of Russia more negatively in security-allied states than in neutral ones.
- “Separating Rate from Composition: A Zero-Loss Decomposition of Electoral Change”. 2025.
Introduces a zero-loss decomposition (RCVD) of electoral change into rate, composition, and volume effects; applied to recent U.S. presidential elections, shows rightward Hispanic shifts in 2024 fully neutralized Democrats’ demographic advantage.

- “Tracking Civic Space in Developing Countries with a High-Quality Corpus of Domestic Media and Transformer Models” with Jeremy Springman, Erik Wibbels, Fatih Serkant Adiguzel, Mateo Villamizar Chaparro, Zung-Ru Lin, Diego Romero, Mahda Soltani, Hanling Su, and Jitender Swami. 2025. Under review at *Political Analysis*.
New monthly measures of civic space activity across 69 developing countries (2012-2024), built from a transformer-based pipeline over 120M+ articles from 350+ domestic media outlets.
- “Foreign Influence by Authoritarian Governments: Introducing New Data and Evidence” with Jeremy Springman, Fatih Serkant Adiguzel, Mateo Villamizar Chaparro, Zung-Ru Lin, Diego Romero and Erik Wibbels. 2025. Under review at the *Journal of Peace Research*.
LLM-based analysis of a 120M-article corpus across 69 aid-receiving countries (2012-2025) documents a previously undetected surge in Russian diplomatic and hard-power activity before the Ukraine invasion.
- “#DataForUkraine: Adapting Social Science Tools for Crisis Response” with Ernesto Calvo, Zung-Ru Lin, Olga Onuch, Graeme Robertson, Jeremy Springman and Erik Wibbels. 2022.
A near real-time, social media-based event detection system built for policymakers during the 2022 invasion of Ukraine, with reflections on crisis-response data infrastructure.

--- Awards & Honors

- Alvin Z. Rubinstein Award for Excellence in Teaching, 2026
- Penn Grad Talks, Social Science Winner, 2026 (*competitive university-wide research communication competition*)
- Steven B. Feirson Graduate Fellow, 2023-2024
- Forever Duke Student Leadership Award, 2019
- Duke Economics Student Leadership Award, 2019

--- Policy Reports

Prepared for USAID’s INSPIRES Consortium and available at web.sas.upenn.edu/mlp-devlab.

- “Violence Against Journalists and Reporting on Civic Space”, with Erik Wibbels 2023
- “Technical Report on the Production of Civil Space and RAI Event Count Data” with Jeremy Springman and Erik Wibbels, 2021
- “Legal Changes and Protest: Evidence from High-Frequency Data” with Jeremy Springman and Erik Wibbels, 2021
- “What Predicts Changes in Civic Space? Evidence from Forecasts in Serbia” with Jeremy Springman and Erik Wibbels, 2021

Research Experience

Research Manager, *DEVLAB@Duke Machine Learning for Peace*, Durham, NC, 2020-2022

- Promoted to Research Manager from Research Project Supervisor; managed a research team of approximately 10 (postdocs, PhD students, data scientists, and RAs) producing data infrastructure and forecasting outputs for the USAID-funded INSPIRES Consortium on civic space resilience.
- Led training data construction, codebook development, and validation for transformer-based classification pipelines translating multilingual news articles and coding them into civic space and foreign influence categories; supervised pipeline engineering and model development in Python.
- Produced quarterly forecasts of civic space activity (protests, arrests, civil society restrictions) across 69 developing countries, delivered to USAID missions and INSPIRES Consortium partners; led annual policy report production.
- Built original public-facing Shiny dashboard for live civic space data, still maintained at web.sas.upenn.edu/mlp-devlab.

Research Intern, Division for Social Policy and Development, *United Nations*, New York, NY, 2018

- Conducted data analysis in R on cross-national disability metrics across 10 developing countries; contributions used in the UN Flagship Report on Disability and Development.
- Supported organization of the 11th Conference of State Parties to the Convention on the Rights of Persons with Disabilities (CRPD), including research, briefings, and side-event documentation.

Conference Presentations

- 2026 *The Signaling Gap*. NEWEPS-27 (accepted).
- 2026 *The Signaling Gap*. APSA Annual Meeting.
- 2026 *The Signaling Gap*. Society for Political Methodology (PolMeth) Annual Meeting.
- 2026 *The Signaling Gap*. NLP Tutorial, SICSS-Penn (invited).
- 2025 *Separating Rate from Composition*. PolMeth Annual Meeting.
- 2025 *Foreign Influence by Authoritarian Governments*. Northeastern Political Science Association.
- 2025 *Separating Rate from Composition*. Empirical Studies of Conflict (ESOC).

Workshop Presentations

- 2026 *The Signaling Gap*. UPenn International Relations Workshop.
- 2026 *The Signaling Gap*. UPenn PDRI Workshop.
- 2026 *The Signaling Gap*. UPenn MEDIATED Workshop.
- 2025 *The Signaling Gap*. UPenn PDRI Workshop.
- 2025 *The Signaling Gap*. UPenn International Relations Workshop.
- 2025 *Tracking Civic Space*. UPenn PDRI Workshop.
- 2024 *Separating Rate from Composition*. UPenn PDRI Workshop.

Teaching

- CETLI Teaching Certificate, University of Pennsylvania, 2026
- STAT 5710: Modern Data Mining, Head TA 2026
- GAFL 5320: IDev Policy Research Shop, TA 2026
- Penn Political Science Ph.D. Math Camp (sole instructor), 2023, 2025
- PSCI 6800: Advanced Statistical Analysis, TA 2023, 2025
- Wharton Data Science Academy, Teaching Fellow, 2025
- PSCI 1408: War, Strategy and Politics, TA 2025
- PSCI 1800: Introduction to Data Science, TA 2024

Service

- Editorial Assistant, Political Analysis 2025-2026
- Reviewer, Political Analysis, Sociological Methods and Research, and the Journal of Conflict Resolution
- Political Science Graduate Student Organization Co-Chair, 2023-2024, 2025-2026
- SASGov Student Representative, 2026
- Professional and Graduate Student Council Representative, Duke University, 2018-2019
- Master of Economics Student Council, Duke University, 2018-2019

Other Professional Experience

Financial Analyst II, *Tempur Sealy International*, Lexington, KY, 2013-2017

- Promoted from Financial Analyst I to Financial Analyst II
- Developed ad hoc models and analyses to support various functions within the business, providing support to sales, operations, marketing, accounting and other various finance teams using SQL and Oracle, including introducing new methods of analysis such as elasticity modeling
- Revamped forecast/budgeting processes to remove at minimum 20 hours of monthly work through improving both methods of analysis and applying greater Excel functionality reducing time spent on low value forecasting
- Finance lead on negotiations to launch Tempur branded products on Amazon.com, responsible for pricing, business cases and marketing plans

Research & Technical Skills

Methods	Causal inference, quasi-experimental design, sensitivity analysis, forecasting, spatial analysis, NLP and text-as-data, transformer models, LLM applications, sentiment analysis
Languages & Tools	R, Python, SQL, Git, LaTeX, Quarto
Selected R Libraries	tidyverse, data.table, fixest, estimatr, MatchIt, sensemakr, sf, ggplot2, patchwork, modelsummary

Selected pandas, numpy, scikit-learn, xgboost, seaborn, matplotlib
Python
Libraries